

RFID Based Smart Electronic Voting System using ESP32 with Duplicate Prevention, Session Control, Logic Gate Integration, and Accountability Tracking

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Abstract-This project presents the design and implementation of a smart electronic voting system using ESP32 and RFID technology. The main objective of the system is to ensure secure, transparent, and reliable voting by allowing only one vote per RFID card and preventing duplicate voting attempts. A session-based control mechanism is implemented to manage voting duration effectively and avoid unauthorized access. The system uses multiple LED indicators and a buzzer to provide real-time feedback to users during the voting process. Initially, the system was tested through simulation using the Wokwi platform, which helped identify logical errors and improve system design. After that, a hardware prototype was developed using breadboard and PCB techniques. During implementation, several practical issues such as incorrect USB cables, LED polarity errors, and wiring faults were identified and solved using debugging tools like a multimeter. The system demonstrates low cost, scalability, and real-world applicability. It can be used in institutions or voting environments where accountability and transparency are required. Overall, the project provides a reliable and efficient solution for modern electronic voting systems.

Index Terms - RFID, ESP32, Electronic Voting, Embedded System, Logic Gates

I. INTRODUCTION

IN recent years, electronic voting systems have gained significant importance due to their ability to improve transparency, efficiency, and accuracy in decision-making processes. Traditional voting systems, such as paper-based ballots or manual counting, often suffer from issues like human error, lack of traceability, time consumption, and potential manipulation. These limitations create a need for a more secure and reliable voting mechanism.

With the advancement of embedded systems and identification technologies, RFID (Radio Frequency Identification) has emerged as an effective solution for user authentication. Each RFID card contains a unique identification number, which can be used to ensure that each individual is allowed to vote only

once. This helps eliminate duplicate voting and enhances system integrity.

In many real-world scenarios, such as institutional elections or parliamentary voting systems, decisions are sometimes influenced by external pressure rather than individual accountability. This reduces transparency and trust in the system. Therefore, there is a strong motivation to develop a voting system where each vote is uniquely linked to an individual and properly recorded.

This project is motivated by the need to design a low-cost, reliable, and secure electronic voting system using ESP32 and RFID technology. The system aims to ensure one-person-one-vote policy, provide real-time feedback, and improve accountability. Additionally, it offers hands-on experience in digital logic implementation, embedded system design, and hardware troubleshooting.

The main objectives of this project are listed below:

- To design and develop a smart electronic voting system using ESP32 microcontroller
- To implement RFID-based authentication to ensure secure user identification
- To prevent duplicate voting using logical conditions and system validation
- To implement session control to restrict voting within a specific time frame
- To provide real-time feedback using LEDs and buzzer for system status indication
- To validate system performance through simulation and hardware implementation

This report is organized into several sections to provide a clear understanding of the project. Section II presents the literature review, which discusses existing research related to electronic voting systems and RFID-based authentication. Section III describes the methodology and modeling of the system, including working principles, digital logic implementation, and component details. Section IV presents the results and discussion, including simulation outcomes, hardware testing results, comparisons, cost analysis, and limitations. Finally, Section V concludes the report and suggests possible future improvements for enhancing system functionality and scalability.

II. LITERATURE REVIEW

Electronic voting systems have been widely researched in recent years to improve transparency, security, and efficiency in election processes. Traditional voting systems often face challenges such as voter impersonation, manual errors, and delayed result processing. Modern research focuses on integrating advanced technologies like RFID, IoT, and encryption to overcome these limitations.

A study by Kumar et al. (2024) analyzed different types of electronic voting machines used worldwide and categorized them into practical and theoretical systems. The study highlights that modern voting technologies aim to improve accessibility, transparency, and system reliability through digital transformation.

Research conducted by Achammal et al. (2021) proposed an RFID-based smart electronic voting system using Arduino. The system ensures secure authentication by using RFID tags and prevents electoral fraud by allowing only authorized users to vote once.

Another recent work by Adewumi (2025) introduced an IoT-based RFID voting system that integrates Firebase and real-time data processing. The study emphasizes secure authentication, prevention of multiple voting, and improved user experience through modern software and hardware integration.

A more advanced approach was presented by Arinze et al. (2025), where an RFID-enabled electronic voting framework was developed with encryption and real-time monitoring. The system uses secure UID verification and AES encryption to ensure data integrity, privacy, and transparency in the voting process.

In addition, Antony et al. (2026) proposed a touchless RFID-based voting system using ESP32 and ultrasonic sensors. This system improves both hygiene and efficiency while ensuring accurate voter authentication and eliminating duplicate voting through cloud-based verification.

Furthermore, recent research on blockchain-based voting systems (Kim et al., 2021) shows that encryption techniques such as homomorphic encryption and blockchain can significantly enhance data security, privacy, and trustworthiness in electronic voting systems.

From the above studies, it is evident that combining RFID authentication with embedded systems like ESP32 can significantly improve voting security, prevent duplicate voting, and enhance system efficiency. However, many existing systems still face challenges related to scalability, cost, and real-world deployment. Therefore, this project focuses on developing a low-cost, reliable, and secure RFID-based voting system with session control and real-time feedback.

III. METHODOLOGY AND MODELING

A. Introduction

This section describes the design and implementation of the proposed RFID-based electronic voting system. The system integrates RFID technology with an ESP32 microcontroller to ensure secure authentication and controlled voting. Logical conditions are applied to validate user input, prevent duplicate voting, and manage voting sessions. The methodology includes system design, working principle, component selection, and experimental setup.

B. Working Principle of the Proposed Project

The system operates based on RFID authentication and logic-based decision making. First, a user scans an RFID card, which contains a unique identification number (UID). The ESP32 reads this UID and checks whether the card is valid and whether the voting session is active.

If both conditions are satisfied, the user is allowed to cast a vote using push buttons. The system ensures that each RFID card can be used only once, preventing duplicate voting. Logical operations similar to AND and OR gates are used to control system behavior.

- AND logic ensures that voting is allowed only when both conditions (valid card and active session) are true.
- OR logic activates the event indicator when any vote (YES/NO) is cast.
- Duplicate detection logic prevents repeated voting attempts.

C. Description of the Components

The system provides real-time feedback through LEDs and a buzzer to indicate successful voting, duplicate attempts, or system status.

The system uses the following components:

- ESP32 Dev Board: Acts as the main controller, processes inputs, and controls outputs.
- RFID RC522 Module: Used for user authentication by reading RFID cards.
- RFID Cards/Tags: Contain unique IDs for each voter.
- LEDs: Indicate system status such as valid vote, duplicate attempt, or system activity.
- Resistors (150Ω): Protect LEDs by limiting current.
- Push Buttons: Used to cast votes (e.g., YES/NO).
- Buzzer: Provides audio feedback for system events.
- Breadboard/PCB: Used for circuit assembly and prototyping.
- Jumper Wires: Used for electrical connections.
- Multimeter: Used for testing and debugging the circuit.

D. Experimental Setup

The system is implemented using a layered architecture

consisting of input, processing, and output units.

- Input Layer: RFID module and push buttons are used to collect user input.
- Processing Layer: ESP32 processes the input data and applies logical conditions.
- Output Layer: LEDs and buzzer provide visual and audio feedback.

During setup, all components are connected with a common ground, and the RFID module is powered using 3.3V. LEDs are connected through resistors to prevent damage. The system is first tested through simulation using the Wokwi platform, followed by hardware implementation on a breadboard and PCB.

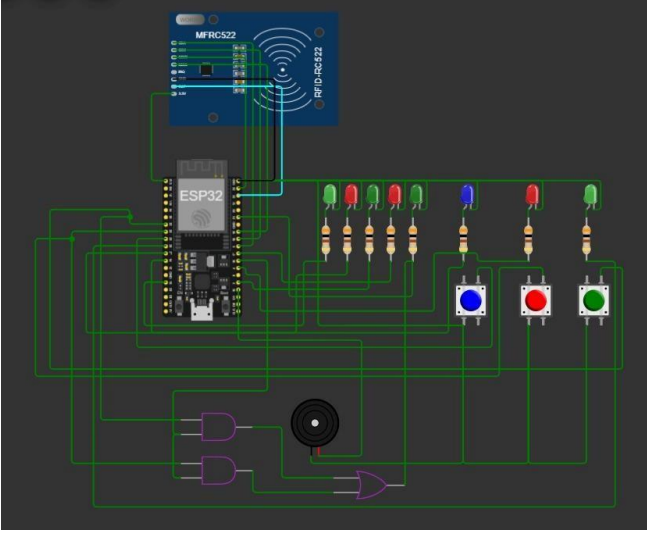


Fig. 1. System Circuit Diagram

IV. RESULTS AND DISCUSSION

A. Simulation Analysis

The proposed system was first tested using the Wokwi simulation platform to verify logical operations and system behavior before hardware implementation. The simulation included RFID input handling, button interactions, LED indications, and session control logic.

Table I Simulation Results:

Test Case	Result
Button Input	Working
Duplicate Check	Successful
Session Timeout	Working

The simulation helped identify logical errors and reduced hardware implementation issues. It confirmed that the system correctly validates users, prevents duplicate voting, and manages session timing effectively.

B. Experimental Results

After successful simulation, the system was implemented on hardware using ESP32, RFID module, LEDs, and other components. The system was tested under different conditions to evaluate its performance.

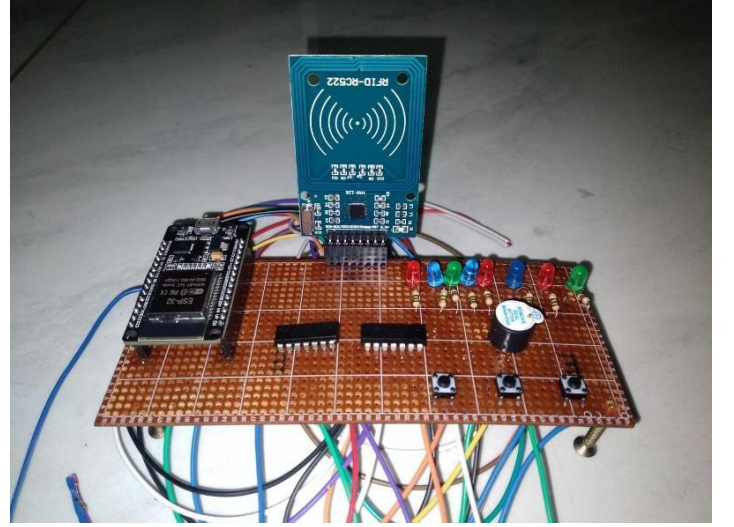


Fig. 2. Hardware Implementation of Voting System Initial State

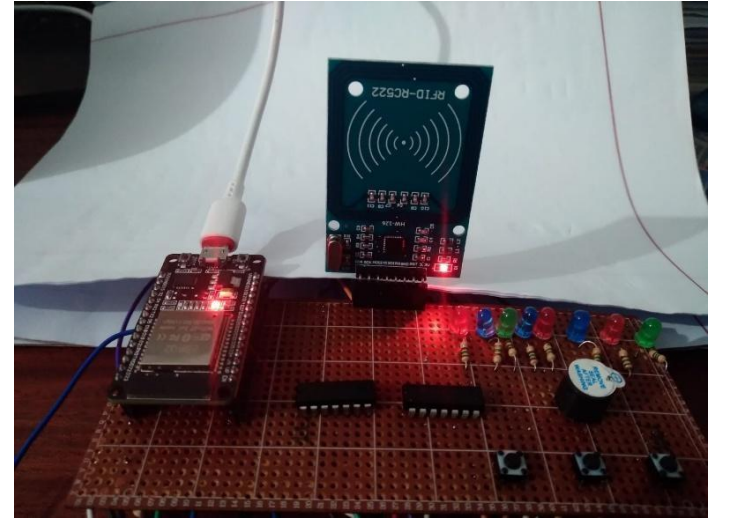


Fig. 3. Hardware Implementation Showing Active Voting State with LED Indicators

Table II Hardware Testing Results:

Test	Outcome
RFID Scan	Success
Duplicate Detection	Success
LED Output	Correct
Session Control	Accurate

During testing, some practical issues such as incorrect USB cable, LED polarity errors, and wiring problems were identified and resolved. The final system operated correctly and provided real-time feedback.

C. Comparison between Simulation and Experimental Results

The simulation and experimental results were compared to evaluate system performance. The simulation provided expected logical behavior under ideal conditions. In contrast, the hardware implementation revealed real-world challenges such as wiring errors and component limitations.

After debugging, both results showed consistent behavior. The system successfully validated users, prevented duplicate voting, and maintained accurate session control in both simulation and real implementation.

D. Cost Analysis

The proposed system is cost-effective and suitable for practical use. The main components such as ESP32 and RFID module are relatively inexpensive, and other components like LEDs, resistors, and buttons are low-cost.

Overall, the system can be implemented with a low budget, making it suitable for institutional and small-scale voting applications.

E. Limitations in the Project

Despite successful implementation, the system has some limitations:

- No cloud or database storage for vote records
- Limited scalability for large-scale elections
- Requires manual debugging and setup
- No advanced security features like encryption or biometric verification

V. CONCLUSION AND FUTURE ENDEAVORS

A. Conclusion

The proposed RFID-based smart electronic voting system using ESP32 has been successfully designed and implemented. The system ensures secure and reliable voting by allowing only one vote per RFID card and effectively preventing duplicate voting attempts. The integration of session control further enhances system security by restricting voting within a defined time period.

Both simulation and hardware implementation confirmed that the system performs accurately under different conditions. The use of LEDs and buzzer provides real-time feedback, making the system user-friendly and easy to operate. Additionally, the project provided practical experience in embedded systems, digital logic implementation, circuit design, and hardware debugging.

Overall, the system is cost-effective, efficient, and suitable for small-scale voting applications where transparency and accountability are important.

B. Future Endeavors

Although the system performs effectively, several improvements can be made in the future to enhance its functionality and scalability:

- Integration of a cloud-based database to store voting records securely
- Development of a web-based monitoring system for real-time result tracking
- Addition of an OLED or LCD display for better user interaction
- Implementation of biometric authentication (fingerprint/face recognition) for higher security
- Use of encryption techniques to protect data transmission and prevent unauthorized access
- Scaling the system for large-scale applications such as institutional or national-level voting

These improvements will make the system more robust, secure, and suitable for real-world deployment.

VI. REFERENCES

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